

Putative role of dietary trace elements in coronary heart disease and cancer

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(Accepted 15 May 1994)

Abstract: Relatively little attention has been given to the role of dietary trace elements in oxidative processes or in the aetiologies of chronic disease processes. Iron and copper are pro-oxidants *in vitro*, but there is now compelling evidence that adequate body copper status is required to maintain antioxidant defences *in vivo*. Epidemiological evidence linking measures of high iron nutritional status with both coronary heart disease (CHD) and cancer is accumulating, although there are few mechanisms implicating iron in these disease processes apart from acting as a pro-oxidant. In contrast, low copper nutritional status may produce pro-oxidant effects and experimental evidence, especially from animal models of CHD, suggests that copper has an involvement in disease mechanisms which is much wider than simply an involvement in maintaining oxidant/antioxidant balance. Zinc is considered to have antioxidant effects *in vivo* but the role of zinc as an antioxidant, or in CHD and cancer processes, is presently unclear. Although selenium has for some time been recognised as an antioxidant nutrient, epidemiological data gathered to date linking this trace element with either CHD or cancer are inconsistent.

Key words: Heart diseases. Neoplasms. Nutrition. Trace elements.

Introduction

The major nutritional emphasis to date in attempts to understand the aetiologies of coronary heart disease (CHD) and cancer has been towards establishing links with dietary fat. Although the dietary fat hypothesis remains the predominant research focus, increasing attention is being given to dietary micronutrients—particularly those that might protect against oxidant damage.

Oxidant damage has been implicated in the pathology of a number of human diseases including CHD and cancer.¹ This damage can arise through oxidative processes mediated by reactive oxygen species (ROS) such as superoxide radical ($O_2^{\cdot-}$), hydroxyl radical ($\cdot OH$) and oxygen centred radicals of organic compounds ($ROO\cdot$, peroxy and $RO\cdot$,

alkoxyl), together with other non-radical reactive compounds, hydrogen peroxide (H_2O_2) and singlet oxygen (1O_2). Important biological defence systems have evolved to limit inappropriate exposure to ROS. For conceptual convenience, antioxidant defences can be classified into three main types on the basis of temporal sequence of action.¹ The primary or preventive antioxidants limit the initial formation of oxygen-centred radicals of organic compounds. The latter ROS are intermediates of peroxidative chain reactions. Secondary scavenging or chain-breaking antioxidants trap these intermediate ROS directly and thus break the chain reaction. Although each of these defence systems requires dietary micronutrients for optimisation, most research effort on antioxidants has centred on attempts to elucidate the protective roles of the chain-breaking (or secondary) antioxidants,

namely vitamin E, vitamin C and the carotenoids. Selenium has also received some attention as this trace element can act, through the selenium-dependent glutathione-peroxidase (GSH-Px) enzymic activity, as a chain-breaking antioxidant in concert with vitamin E in detoxifying organic peroxy radicals and related compounds. The GSH-Px enzyme can also act as a primary antioxidant in the conversion of hydrogen peroxide to water. Much of the primary antioxidant defence, however, is concerned with sequestering transition metal (copper and iron) ions in forms which are incapable of catalysing the conversion of superoxide (the predominant initial ROS formed when oxygen is 'activated' within aerobic organisms) and hydrogen peroxide into more damaging species (e.g. the hydroxyl radical). The latter ROS are considered to be the initiating factors in lipid peroxidation and other chain reactions involving oxidation of macromolecules. A number of trace elements, therefore, are implicated in oxidative processes and antioxidant defences in the body but, in general, trace elements have received much less attention than the antioxidant vitamins in the aetiologies of chronic disease processes.

This review approaches the aetiologies of CHD and cancer from the oxidant/antioxidant viewpoint, and concentrates on the trace elements that are quantitatively the most important – iron, zinc and copper, together with selenium. This is certainly not an exhaustive list of the trace elements that have been implicated in either disease. For example chromium,² vanadium³ and manganese⁴ have each been associated with glucose intolerance, a risk factor for CHD, while iodine (or at least thyroid hormone imbalance) has been associated with both CHD^{5,6} and cancer⁷. In addition, manganese is involved in antioxidant defence as a component of the mitochondrial superoxide dismutase, which dismutates the potentially damaging reactive oxygen species, superoxide anion, to hydrogen peroxide and water.⁴ Another trace element in the transition series, molybdenum, has been shown to inhibit mammary cancer in rats, and molybdenum deficiency may be involved in the aetiology of human oesophageal cancer.⁸

Coronary heart disease

The major processes involved in CHD are atherosclerosis, thrombosis, and arterial and cardiac dysfunction. As with other postulated mechanisms associated with CHD, the role of oxidative damage in atherosclerosis has received most interest. The study

of atherosclerosis, in turn, has been dominated by attempts to understand the mechanisms whereby the major cell type of the atherosclerotic lesion, the monocyte-derived macrophage, accumulates cholesterol from low-density lipoproteins (LDL). These cells have a well-regulated control on the uptake of cholesterol from LDL. Apparently, it is only when the LDL are modified by oxidation (or other means) that they are scavenged avidly, transforming the macrophage into a very reactive cell capable of producing growth promoters which induce the replication of neighbouring smooth muscle cells. Eventually the macrophages, engorged with oxidised LDL, appear as foam cells within the progressive thickening or plaque in the arterial wall.⁹ Oxidised-LDL, unlike natural LDL, also have the potential to attract monocytes into, and to prevent the movement of the derived macrophages out of, the intimal layer of the arterial wall. Moreover oxidised-LDL are immunogenetic and cytotoxic. Modified lipoproteins, therefore, may initiate and further amplify inflammatory stress response, and may lead to endothelial cell damage and disruption of endothelial function.¹⁰

These processes (and oxidised LDL *per se*) could be responsible for: (a) adversely altering coagulation pathways by activating platelets, increasing thromboplastin activity, or by stimulating tissue factor secretion; and (b) adversely altering vasomotor properties of coronary arteries by interfering with endothelial-dependent relaxing factor (i.e. nitric oxide, another ROS)-mediated vasodilation. Perturbations in eicosanoid production may also play a role in these deleterious changes in blood clotting and arterial function. Whatever the mechanisms involved in these changes, it is clear that a potential role for increased oxidant stress and oxidised LDL in the pathogenesis of CHD extends beyond, and is certainly not confined to, atherosclerosis.

Iron

There is currently much debate on how ROS might be implicated in LDL oxidation. Although the nature of the initial oxidising species in *in vitro* culture is controversial, lipid peroxidation – and presumably the oxidation of other biomolecules – is absolutely dependent on the presence of a transition metal catalyst such as iron or copper. Yet under normal physiological conditions *in vivo*, the concentrations of both iron and copper in forms capable of catalysing

oxidation of biomolecules are extremely low or negligible. Indeed, much of the primary antioxidant defence is to ensure that these transition metals are tightly regulated and controlled. One possibility is that superoxide generation *in vivo*, or compounds capable of undergoing single electron oxidation/reduction reactions, may mobilise iron from ferritin, the major iron-storage protein.¹¹ Haemoglobin release after haemolysis and internal bleeding, or myoglobin release after myocardial infarction, may also contribute to catalytic forms of iron.¹²

There is now clear evidence indicating iron-induced oxidative damage in iron overload conditions, but there is no evidence of higher rates of CHD in subjects homozygous or heterozygous for the iron-loading genetic disorder, haemochromatosis.¹³ Sullivan¹⁴ has argued that the greater incidence of CHD in men and post-menopausal women, compared with the incidence in pre-menopausal women, is due to higher levels of stored iron in the former two groups. He supports his argument by observations of (a) myocardial failure (but not necessarily CHD) in iron storage diseases, (b) accumulation of stored iron with age in men, and (c) accumulation of stored iron after menopause in women to levels found in men. Other observations, such as the apparent association of iron depletion with low CHD risk in people eating Third World diets and subject to parasite loads, fish consumption, blood donation, or long-term use of aspirin or cholestyramine, have also been used by Sullivan¹⁵ to support his hypothesis. Certainly, major end-points in haemochromatosis are diabetes¹⁶ (an important risk factor for CHD), cardiac arrhythmias and congestive heart failure.

There are some epidemiological observations from prospective¹⁷ and case-control studies¹⁸ which suggest that moderately high haematocrit and haemoglobin levels are risk factors for CHD, while more recently Salonen and co-workers,¹⁹ in a prospective study in Finland, found that raised serum ferritin was a risk factor for CHD. Lauffer²⁰ has also provided ecological data correlating 'liver-iron' serum cholesterol product with CHD mortality in men and women from 11 countries. These observations are supported by animal studies where high dietary iron levels can increase serum cholesterol.²¹ Magnusson *et al.*,²² however, did not find raised serum ferritin (and therefore iron stores) to be a risk factor for CHD in a small prospective Icelandic study, but rather low iron-binding capacity (or transferrin) was an independent risk factor in this population.

Notwithstanding these data, there are few mechanistic relationships which can be cited to link

moderately high iron status with the other major CHD processes of thrombosis or arterial dysfunction, whereas raised serum ferritin levels in population studies could be interpreted as a consequence of an acute phase reaction in response to a developing atheroma, rather than as a marker of iron storage. Consequently, with respect to the latter, disproportionately elevated serum ferritin relative to transferrin saturation could be indicative of chronic inflammation in men and older women in Northern Ireland,²³ rather than the high serum ferritin being interpreted as a causative factor in the current high CHD rates in this population. The suggestion, therefore, that high serum ferritin levels indicate high iron stores which, in turn, are causally related to CHD, should be interpreted with caution until more information is available on the oxidative or other effects of a moderately high body iron status within the 'normal' range found in a given population.

Zinc

The physiological role of zinc as an antioxidant is presently unclear.²⁴ It may protect sulphhydryl groups against oxidation, inhibit the production of ROS by transition metals, and stabilise cell membranes. Another possible mechanism whereby zinc might act as an antioxidant is through the induction of metallothionein – a ROS scavenger.²⁵ Poor zinc status can inhibit the synthesis of metallothionein and can negate the protective effects of the acute phase response to carbon tetrachloride ROS-mediated hepatotoxicity.²⁶

Although evidence is accumulating that zinc deficiency compromises antioxidant defences, there is no clearly defined role for low zinc status in CHD. Hyperzincuria is a feature of non-insulin dependent diabetes mellitus, and may eventually lead to zinc deficiency.²⁷ Alternatively, zinc loss may simply accompany lean body tissue losses in diabetes mellitus.⁴ It is well known, however, that excess zinc can inhibit copper absorption, which in turn may compromise antioxidant defences.²⁸

Copper

Many transition metal ions (and complexes) in lower oxidation states can act, in the presence of hydrogen peroxide, as 'Fenton-like' reagents by producing the highly reactive hydroxyl radical.²⁹ The latter ROS is considered by many as an essential initiator of oxidative

reactions involving biomolecules. Apparently the mechanisms for *in vitro* oxidation of LDL differ, depending on which of these metal ions are used to catalyse the oxidative process.³⁰ Although copper is the usual metal ion used to induce oxidation of lipoproteins *in vitro* by many laboratories,³¹ there is little (if any) experimental evidence to indicate that copper might be involved in similar events *in vivo*. Iron, in contrast, appears to be a much more likely catalyst of oxidative processes *in vivo*. Iron can be mobilised from ferritin by superoxide radical, and limiting the initiation of oxidative damage may depend on the enzymic activities of the copper-containing proteins superoxide dismutase (Cu,Zn-SOD copper but not zinc dependent), which dismutates superoxide, and caeruloplasmin, which reincorporates the released iron from ferritin. Moreover, copper-amino acid complexes may prevent iron release from ferritin (reviewed by Reif¹¹).

Copper, therefore, may play an antioxidant, rather than pro-oxidant, role *in vivo*, and there is now appreciable evidence arising from *in vivo* studies to indicate an antioxidant role for copper through involvement in maintaining antioxidant defences.²⁸ The susceptibility of both very low-density lipoprotein (VLDL) and LDL to *in vitro* copper-induced oxidation has recently been shown to be increased in copper-deficient rats.³² Further recent evidence indicates that the hypercholesterolaemia of copper deficiency, first reported by Klevay³³ over 20 years ago, is regulated by the antioxidant glutathione.³⁴ Moreover, feeding low copper diets to rats impairs vascular function and increases cardiac³² and aortic³⁵ lipid peroxides; the latter increase being associated with depressed aortic Cu,Zn-SOD activity. The activity of Cu,Zn-SOD was also decreased in both aneurysmal and occluded aortic tissue from humans,³⁶ while patients with arterial disease had significantly lower erythrocyte SOD compared with healthy persons.³⁷

Certainly not all of the many cardiovascular changes associated with copper deficiency in animals³⁸ can be attributed to defective antioxidant defences. Increased oxidant stress owing to defective mitochondria and decreased activity of the copper-dependent cytochrome C oxidase activity, may also be involved.³⁹ Whatever the associated mechanisms, partial inhibition of the detrimental cardiovascular effects of copper deficiency were observed after feeding dimethyl sulphoxide⁴⁰ and t-butylhydroquinone,⁴¹ but not after feeding the natural antioxidant vitamin E.⁴² Cardiac integrity, but not cardiac hypertrophy, was maintained

in rats given dietary supplements of the endogenous antioxidant coenzyme Q₁₀.⁴³ Moreover, 'increased oxidant stress *via* sub-maximal aerobic exercise training exacerbated the cardiomyopathy of copper-deficient rats.⁴⁴ Reduced expression of the selenium-dependent glutathione peroxidase,⁴⁵ and heat shock proteins⁴⁶ in copper deficiency, may also implicate oxidant or other stresses in the adverse changes in cardiovascular physiology and cardiac biochemistry, when animals are challenged with a dietary copper deficiency.

The severity of the cardiovascular defects of copper deficiency is dependent upon the type of dietary carbohydrate, gender and animal species (reviewed by O'Dell⁴⁷). Fields *et al.*⁴⁸ have postulated that the severity of some of these defects upon feeding fructose can be explained by anaemia and the well known impairment of iron metabolism in copper deficiency. Dietary lipids,⁴⁹ protein⁵⁰ and amino acids (including histidine,⁵¹ cystine⁵², methionine⁵³ and homocysteine⁵⁴) and other elements⁵⁵ such as zinc, cadmium and molybdenum can also exacerbate cardiovascular and/or biochemical changes associated with copper deficiency.

It is apparent, however, that not all of the diverse cardiac histological, ultrastructural and electrophysiological abnormalities in copper deficiency^{38,56-58} involve impaired antioxidant status or increased oxidant stress. Changes in cardiac catecholamine metabolism⁵⁹⁻⁶¹ through decreased activity of copper-dependent dopamine β -hydroxylase, and disruption of connective tissue integrity though decreased activity of copper-dependent lysyl oxidase,⁶² may also contribute to these cardiac abnormalities. Similarly, reduced activity of the latter enzyme may play a role in vascular changes which accompany copper deficiency. These vascular defects³⁸ include structural changes in the aorta, decreased functional responses to noradrenaline in portal veins and pulmonary arteries and, more recently, reduced endothelium-dependent relaxation in the rat aorta.^{63,64} Additionally, recent studies have reported that a number of major functional changes occur in the microvasculature of copper-deficient rats.⁶⁵ The latter include altered microvascular leakage, thrombus formation and reduced nitric oxide-mediated relaxation in the cremaster muscle microcirculation. All of these effects suggest altered microvascular endothelial cell function. The observed reduction in microvascular dilation is of interest because of its possible relationship to hypertension - another feature

of copper deficiency.⁶⁶ Depressed prostacyclin production by the endothelium, leading to increased platelet aggregation, could explain – together with a defective physiological clot-lysing mechanism⁶⁷ – the thromboses associated with dietary copper deficiency in rodents.

Even though many of these physiological abnormalities have been demonstrated in animal models only, hypercholesterolaemia, decreased glucose tolerance (historically one of the earliest recognised signs of copper deficiency in animals), abnormal blood pressure response to sustained hand grip, and abnormal electrocardiograms have been found in copper depletion experiments with humans.⁵⁵ However, high serum caeruloplasmin (the major copper-containing protein in serum) is a risk factor for myocardial infarction,⁶⁸ but this increase should not be equated with increased body copper status. Serum copper and caeruloplasmin levels are increased as part of the acute-phase process in inflammatory conditions⁶⁹ such as CHD and are thus, contrary to a recent report relating high serum copper with atherogenesis,⁷⁰ not accurate indicators of copper status in human populations.^{71,72} Analogously, even though oxidative mechanisms have been proposed to explain some of the pathologies associated with the copper loading genetic disorder, Wilson's disease,⁷³ there is no evidence of higher rates of CHD in these subjects. Nevertheless, severe copper deficiency has only rarely been documented in humans,⁷⁴ and the question of whether a mild chronic copper deficiency is a significant nutritional problem in human populations remains unanswered. Moreover, long-term marginal copper deficiency in animals can have quite different effects on cardiovascular variables than a severe copper deficiency.⁷⁵

Selenium

Selenium has antioxidant properties *in vivo* through optimising the activities of glutathione peroxidase, which catalyses the conversion of hydrogen peroxide to water and phospholipid hydroperoxide glutathione peroxidase. The latter enzyme appears to act in a chain-breaking antioxidant mode in concert with vitamin E, and this close metabolic relationship may explain the overlapping and synergistic inter-relationships of these two nutrients in the aetiologies of a number of diseases in man and animals.⁷⁶

Keshan disease is a cardiomyopathy which affects children and young women in parts of China. Low

selenium intakes have been shown to be an important factor in the aetiology of this disease.⁷⁷ The evidence from case-control and prospective studies, however, for an association between low selenium status and CHD is inconclusive.^{78,79} One possible reason for the inconsistencies in the literature is that differences in selenium (serum or toenail) status influence risk only below a threshold value. In a recent European case-control study (the EURAMIC study), low toenail selenium was found to be associated with increased risk of myocardial infarction in men.⁸⁰ Risk was greater in smokers and those with low vitamin E (adipose tissue alpha-tocopherol) status.

Mechanisms whereby selenium might interact with vitamin E in protecting against CHD include increased resistance of LDL against oxidative modification, especially among smokers. Selenium also appears to have overlapping roles with vitamin E in regulating prostaglandin synthesis and platelet aggregation.⁸¹ In addition, interactions among copper, selenium and glutathione peroxidase in antioxidant defence have been noted.^{54,82,83} Indeed, copper deficiency can lower rat hepatic levels of glutathione peroxidase mRNA, but not selenium.⁴⁵ Although there are some similarities between the cardiomyopathies caused by copper or selenium deficiencies in rats, differences are also apparent.⁸⁴ The latter workers suggested that cardiac changes associated with both of these trace element deficiencies may be related to mechanisms other than lipid peroxidation.

Cancer

While the causes of cancer have not been clearly elucidated, it is now widely accepted that persistent alterations in DNA – which, following DNA replication, become fixed as mutations – are important mechanistic processes. Most DNA, however, is not expressed in a given cell, and a continuous flux of cancer initiators are required as most mutations have little or no effect on cell function. Moreover, DNA repair mechanisms have been identified in normal healthy individuals and these processes compete effectively to limit the background levels of mutations in DNA.⁸⁵ It is well known that individuals who have a defect or deficiency in DNA repair have increased risk of cancer.

Ames⁸⁶ has suggested that significant oxidative damage occurs *in vivo* as a result of ROS attack, and this contributes to the aetiology of cancer. Indeed, it is becoming increasingly apparent that many genotoxic

agents act through the common mechanism of oxidative damage to DNA.¹ As well as being important initiators of carcinogenic changes *via* oxidative damage, ROS may act as tumour promoters, modulating the expression of genes which regulate cell differentiation and growth and acting synergistically with the initiators.^{87,88} Oxidative processes may also facilitate the activation of latent collagenases to an inactive form needed for metastases.⁸⁹

One of the difficulties in studying the aetiology of cancer is the heterogeneity of the disease. This review will focus on the possible role of trace elements in the aetiologies of the major prevalent cancers in western societies (*i.e.* lung, colon and breast).

Iron

Liver cancer is a common occurrence in iron overload,⁸⁹ and there also appears to be an increased risk of extrahepatic carcinomas in hereditary haemochromatosis.⁹⁰ Moreover, a number of prospective epidemiological studies have found a positive relationship between iron status and cancer risk.⁹¹ Two of these studies involved large cohorts of subjects in the United States.^{92,93} Both studies used total iron-binding capacity (TIBC) or transferrin saturation (rather than the more definitive serum ferritin) as measures of body iron stores.

In the Kaiser Study,⁹² over 175 000 members of a health plan in northern California were followed from 1964 to 1973. Baseline medical examination and blood measurements were taken during this period, and the population was then followed through 1980 and incident cases of cancer reported. Cases occurring within two years of the blood tests were excluded from the analyses. Adjustment was made for age and smoking, which can affect iron status measurements in population studies.⁹⁴ Women, but not men, who reported recent iron supplement use had a lower risk (relative risk=0.6) of lung cancer than those who did not. Furthermore anaemia (haemoglobin < 12 g/l) was associated with lower risk (relative risk=0.61), while TIBC was inversely related to risk (relative risk=0.41), comparing highest with lowest quintile of lung cancer in women but not in men.

In the NHANES I Study in the United States, Stevens *et al.*⁹³ compared transferrin saturation and TIBC in over 14 000 subjects who developed cancer and those who did not. Cases occurring within four years of medical examination and blood measurements were not included in the analyses, and again

adjustment was made for age and smoking. In contrast with the Kaiser Study, cancer (all sites) risk was weakly related to the iron status measurements in men but not in women. Both of these studies, however, used inappropriate measures of iron stores. The possibility that the iron status measurements were affected by latent cancer or as yet undiagnosed cancer also can not be ignored, even though two or four year 'cancer-free' intervals were employed from the time of blood sampling. Indeed, Yip *et al.*⁹⁵ found no association of transferrin saturation with subsequent risk of cancer on reanalysis of the NHANES data, where initial health status was controlled for more carefully and/or a longer follow-up period was employed. The previously reported association by Stevens *et al.*,⁹³ therefore, appears to be an artefact of the effect of pre-existing disease on transferrin saturation.

In spite of the problems in interpretation of the data reported from epidemiological studies to date, there is considerable evidence from clinical and experimental human cancer,⁹⁶ and from animal studies,⁹⁷ that iron status plays a role in neoplasia. Although iron is an essential component in DNA synthesis and in respiratory and oxidative metabolism through iron-dependent enzymes such as mitochondrial cytochromes, cytochrome P-450, catalase, and ribonucleotide reductase, iron-DNA interactions can stimulate gene mutation,⁹⁸ and *in vitro* exposure to elevated levels of iron results in oxidative DNA damage, particularly single-strand breaks.^{99,100} Increased plasma lipid peroxidation and low antioxidant vitamin status appear to be associated with hereditary haemochromatosis,¹⁰¹ but experimental chronic iron overload in rats, while inducing hepatic lipid peroxidation, does not cause a corresponding significant increase in DNA damage.¹⁰²

Apart from a putative role in increasing DNA oxidant damage, iron may also promote cancer cell growth. Weinberg¹⁰³ has argued that the ability of the host to withhold iron from cancer cells is an important defence strategy. Further, the antineoplastic activity of a considerable variety of chemicals and of radiation is modulated by iron, whereas lowering the exposure of humans to iron and the suppression of the ability of neoplastic cells to assimilate iron might be important in the prevention and therapy of malignant disease. In any event iron status is affected by a large number of variables and is only weakly associated with dietary iron levels.¹⁰⁴ Dietary manoeuvres such as iron and/or vitamin C supplements and iron fortification of foods, however, can markedly increase iron status.

It seems prudent, therefore, to limit iron supplements or food fortification until further evidence is available. Iron supplements also adversely affect the absorption of other trace elements including zinc and copper.^{82,105}

Zinc

Zinc has important roles in both DNA and chromatin structure and function.^{106,107} The frequency of single-strand DNA breaks in liver are increased in zinc-deficient rats, compared with controls.¹⁰⁸ Possible mechanisms put forward to explain this observation were changes in chromatin structure,¹⁰⁹ impaired activity of DNA repair enzymes with zinc 'finger' motifs¹¹⁰ or decreased protection against oxidant damage. Decreased induction of metallothionein in zinc deficiency might be important with respect to the latter postulated mechanism, as it has been speculated that metallothionein acts as a ROS scavenger or as a zinc donor for DNA repair enzymes.²⁵

Although lowered plasma or leucocyte zinc is a feature of various cancers,^{111,112} there is little evidence for zinc deficiency in patients with active disease. The lowered plasma zinc level is most probably a result of the well known response of the body to inflammatory processes and (like iron or copper) changes in blood levels associated with latent or active cancer should be interpreted with caution. The involvement of zinc, however, in DNA processes and in the maintenance of the immune system¹¹³ suggests that more information is required on the role of this trace element in human cancers.

Copper

The involvement of copper in the cancer process *via* pro-oxidant or antioxidant effects is presently unclear. There is altered copper metabolism with elevated serum copper in cancer patients,¹¹⁴ while copper complexes have been shown to have anti-cancer, anti-carcinogenic and anti-mutagenic properties.¹¹⁵ Owing to the acute phase response, elevated serum copper or caeruloplasmin in cancer does not necessarily imply increased body copper status.^{69,71}

Diets rich in fruit and vegetables have consistently been associated with lower cancer incidence, and this association has usually been attributed to increased intake of antioxidant vitamins.^{116,117} In contrast, relatively little attention has been paid to fruits and vegetables as dietary sources of trace elements. This

is understandable as they are generally poor sources of iron, zinc and selenium. Fruit and vegetables, however, can contribute up to 30% of total copper intakes in some Western populations.¹¹⁸

Selenium

Most of the animal studies have provided evidence for an anti-cancer and anti-mutagenic role for selenium in a variety of cancer models.^{119,120} It is probable, however, that many of the protective effects may have resulted from the pharmacological doses generally used in these trials. The mechanisms whereby selenium may play a protective role seem to extend beyond the antioxidant effect of maintaining optimum activities of glutathione peroxidases, and thereby protecting cellular membranes against oxidative damage. Other postulated protective mechanisms include reducing the mutagenicity of cancer-causing chemicals, affecting carcinogen metabolism, inhibiting DNA synthesis and cellular proliferation, and stimulating the immune system.^{121,122}

Data on selenium and cancer from epidemiological studies, however, are inconsistent. Much of the evidence for a protective effect of selenium comes from inter-population studies which have correlated either dietary or blood levels of selenium with cancer incidence. Cohort studies using pre-diagnostic serum or toenail selenium levels do not give any strong support for an association between selenium status and cancer at any site including lung, colon and breast.^{123,124}

Conclusion

This review has concentrated on the potential oxidant/antioxidant roles of trace elements in the aetiology of CHD and cancer. Iron is considered to be a pro-oxidant, and there is some evidence to suggest a weak link between higher iron status and increased risk of either CHD or cancer. Zinc may have some antioxidant properties *in vivo*, but there are few mechanistic links between zinc metabolism and CHD or cancer processes. Although copper has pro-oxidant effects *in vitro* there is accumulating evidence that copper may act as an antioxidant nutrient *in vivo*. Moreover, there are abundant mechanistic relationships involving copper and the processes associated with CHD in particular. These are largely confined, however, to copper deficiency studies in animal models. A major problem in elucidating roles

for either copper or zinc in human disease is the lack of any adequate index of nutritional status for these trace elements. Selenium has well defined antioxidant properties and has been evaluated in a relatively large number of epidemiological studies. Nevertheless, a protective effect of selenium in either CHD or cancer remains equivocal.

Future research on links with human disease should concentrate on the new concepts which have been modifying the objectives of nutrition research on the trace elements.¹²⁵ These are that: (i) there should be more concern for balanced interactions within the whole diet rather than simply with individual elements; (ii) the concept of the total dose-response curve should eventually replace the division of elements into essential and toxic categories; (iii) the concern for the total health effect of elements should complement the hitherto dominating role of the concept of deficiency in the determination of nutritional requirements. It is important to consider these new concepts when trying to understand the possible role of trace elements in chronic disease processes. In particular, how marginal (sub-clinical) excesses or deficiencies of trace elements over a long period of time can have effects on health, and how these effects can be influenced by interactions with other components within the whole diet.

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